

Pearson's theorem.

Chi-squared goodness-of-fit test is based on a probabilistic result that we will prove in this section.

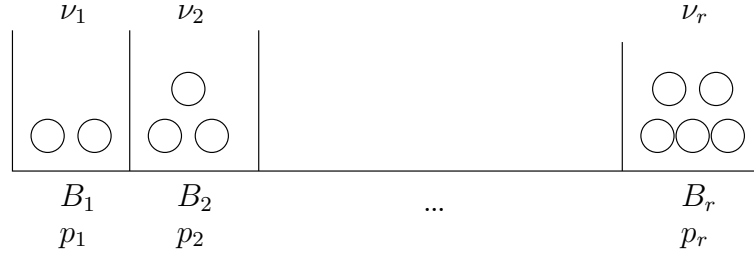


Figure 10.1:

Let us consider r boxes $B_1, \dots, B_r$ and throw n balls $X_1, \dots, X_n$ into these boxes independently of each other with probabilities

$$\mathbb{P}(X_i \in B_1) = p_1, \dots, \mathbb{P}(X_i \in B_r) = p_r,$$

so that

$$p_1 + \dots + p_r = 1.$$

Let ν_j be a number of balls in the j th box:

$$\nu_j = \#\{\text{balls } X_1, \dots, X_n \text{ in the box } B_j\} = \sum_{l=1}^n I(X_l \in B_j).$$

On average, the number of balls in the j th box will be np_j since

$$\mathbb{E}\nu_j = \sum_{l=1}^n \mathbb{E}I(X_l \in B_j) = \sum_{l=1}^n \mathbb{P}(X_l \in B_j) = np_j.$$

We can expect that a random variable ν_j should be close to np_j . For example, we can use a Central Limit Theorem to describe precisely how close ν_j is to np_j . The next result tells us how we can describe the closeness of ν_j to np_j simultaneously for all boxes $j \leq r$. The main difficulty in this Theorem comes from the fact that random variables ν_j for $j \leq r$ are not independent because the total number of balls is fixed

$$\nu_1 + \dots + \nu_r = n.$$

If we know the counts in $n - 1$ boxes we automatically know the count in the last box.

Theorem.(Pearson) *We have that the random variable*

$$\sum_{j=1}^r \frac{(\nu_j - np_j)^2}{np_j} \rightarrow^d \chi_{r-1}^2$$

converges in distribution to χ_{r-1}^2 -distribution with $(r - 1)$ degrees of freedom.

Proof. Let us fix a box B_j . The random variables

$$I(X_1 \in B_j), \dots, I(X_n \in B_j)$$

that indicate whether each observation X_i is in the box B_j or not are i.i.d. with Bernoulli distribution $B(p_j)$ with probability of success

$$\mathbb{E}I(X_1 \in B_j) = \mathbb{P}(X_1 \in B_j) = p_j$$

and variance

$$\text{Var}(I(X_1 \in B_j)) = p_j(1 - p_j).$$

Therefore, by Central Limit Theorem the random variable

$$\begin{aligned} \frac{\nu_j - np_j}{\sqrt{np_j(1 - p_j)}} &= \frac{\sum_{l=1}^n I(X_l \in B_j) - np_j}{\sqrt{np_j(1 - p_j)}} \\ &= \frac{\sum_{l=1}^n I(X_l \in B_j) - n\mathbb{E}}{\sqrt{n\text{Var}}} \rightarrow^d N(0, 1) \end{aligned}$$

converges in distribution to $N(0, 1)$. Therefore, the random variable

$$\frac{\nu_j - np_j}{\sqrt{np_j}} \rightarrow^d \sqrt{1 - p_j}N(0, 1) = N(0, 1 - p_j)$$

converges to normal distribution with variance $1 - p_j$. Let us be a little informal and simply say that

$$\frac{\nu_j - np_j}{\sqrt{np_j}} \rightarrow Z_j$$

where random variable $Z_j \sim N(0, 1 - p_j)$.

We know that each Z_j has distribution $N(0, 1 - p_j)$ but, unfortunately, this does not tell us what the distribution of the sum $\sum Z_j^2$ will be, because as we mentioned above r.v.s ν_j are not independent and their correlation structure will play an important role. To compute the covariance between Z_i and Z_j let us first compute the covariance between

$$\frac{\nu_i - np_i}{\sqrt{np_i}} \text{ and } \frac{\nu_j - np_j}{\sqrt{np_j}}$$

which is equal to

$$\begin{aligned} \mathbb{E} \frac{\nu_i - np_i}{\sqrt{np_i}} \frac{\nu_j - np_j}{\sqrt{np_j}} &= \frac{1}{n\sqrt{p_i p_j}} (\mathbb{E}\nu_i \nu_j - \mathbb{E}\nu_i np_j - \mathbb{E}\nu_j np_i + n^2 p_i p_j) \\ &= \frac{1}{n\sqrt{p_i p_j}} (\mathbb{E}\nu_i \nu_j - np_i np_j - np_j np_i + n^2 p_i p_j) = \frac{1}{n\sqrt{p_i p_j}} (\mathbb{E}\nu_i \nu_j - n^2 p_i p_j). \end{aligned}$$

To compute $\mathbb{E}\nu_i \nu_j$ we will use the fact that one ball cannot be inside two different boxes simultaneously which means that

$$I(X_l \in B_i)I(X_l \in B_j) = 0. \tag{10.0.1}$$

Therefore,

$$\begin{aligned}
\mathbb{E}\nu_i\nu_j &= \mathbb{E}\left(\sum_{l=1}^n I(X_l \in B_i)\right)\left(\sum_{l'=1}^n I(X_{l'} \in B_j)\right) = \mathbb{E}\sum_{l,l'} I(X_l \in B_i)I(X_{l'} \in B_j) \\
&= \underbrace{\mathbb{E}\sum_{l=l'} I(X_l \in B_i)I(X_{l'} \in B_j)}_{\text{this equals to 0 by (10.0.1)}} + \mathbb{E}\sum_{l \neq l'} I(X_l \in B_i)I(X_{l'} \in B_j) \\
&= n(n-1)\mathbb{E}I(X_l \in B_i)\mathbb{E}I(X_{l'} \in B_j) = n(n-1)p_i p_j.
\end{aligned}$$

Therefore, the covariance above is equal to

$$\frac{1}{n\sqrt{p_i p_j}}\left(n(n-1)p_i p_j - n^2 p_i p_j\right) = -\sqrt{p_i p_j}.$$

To summarize, we showed that the random variable

$$\sum_{j=1}^r \frac{(\nu_j - np_j)^2}{np_j} \rightarrow \sum_{j=1}^r Z_j^2.$$

where normal random variables $Z_1, \dots, Z_n$ satisfy

$$\mathbb{E}Z_i^2 = 1 - p_i \text{ and covariance } \mathbb{E}Z_i Z_j = -\sqrt{p_i p_j}.$$

To prove the Theorem it remains to show that this covariance structure of the sequence of (Z_i) implies that their sum of squares has χ_{r-1}^2 -distribution. To show this we will find a different representation for $\sum Z_i^2$.

Let $g_1, \dots, g_r$ be i.i.d. standard normal random variables. Consider two vectors

$$\mathbf{g} = (g_1, \dots, g_r)^T \text{ and } \mathbf{p} = (\sqrt{p_1}, \dots, \sqrt{p_r})^T$$

and consider a vector $\mathbf{g} - (\mathbf{g} \cdot \mathbf{p})\mathbf{p}$, where $\mathbf{g} \cdot \mathbf{p} = g_1\sqrt{p_1} + \dots + g_r\sqrt{p_r}$ is a scalar product of $\mathbf{g}$ and $\mathbf{p}$. We will first prove that

$$\mathbf{g} - (\mathbf{g} \cdot \mathbf{p})\mathbf{p} \text{ has the same joint distribution as } (Z_1, \dots, Z_r). \quad (10.0.2)$$

To show this let us consider two coordinates of the vector $\mathbf{g} - (\mathbf{g} \cdot \mathbf{p})\mathbf{p}$:

$$i^{th} : g_i - \sum_{l=1}^r g_l \sqrt{p_l} \sqrt{p_i} \quad \text{and} \quad j^{th} : g_j - \sum_{l=1}^r g_l \sqrt{p_l} \sqrt{p_j}$$

and compute their covariance:

$$\begin{aligned}
&\mathbb{E}\left(g_i - \sum_{l=1}^r g_l \sqrt{p_l} \sqrt{p_i}\right)\left(g_j - \sum_{l=1}^r g_l \sqrt{p_l} \sqrt{p_j}\right) \\
&= -\sqrt{p_i} \sqrt{p_j} - \sqrt{p_j} \sqrt{p_i} + \sum_{l=1}^n p_l \sqrt{p_i} \sqrt{p_j} = -2\sqrt{p_i p_j} + \sqrt{p_i p_j} = -\sqrt{p_i p_j}.
\end{aligned}$$

Similarly, it is easy to compute that

$$\mathbb{E}\left(g_i - \sum_{l=1}^r g_l \sqrt{p_l} \sqrt{p_i}\right)^2 = 1 - p_i.$$

This proves (10.0.2), which provides us with another way to formulate the convergence, namely, we have

$$\sum_{j=1}^r \left(\frac{\nu_j - np_j}{\sqrt{np_j}} \right)^2 \rightarrow^d |\mathbf{g} - (\mathbf{g} \cdot \mathbf{p})\mathbf{p}|^2.$$

But this vector has a simple geometric interpretation. Since vector $\mathbf{p}$ is a unit vector:

$$|\mathbf{p}|^2 = \sum_{l=1}^r (\sqrt{p_l})^2 = \sum_{l=1}^r p_l = 1,$$

vector $\mathbf{V}_1 = (\mathbf{p} \cdot \mathbf{g})\mathbf{p}$ is the projection of vector $\mathbf{g}$ on the line along $\mathbf{p}$ and, therefore, vector $\mathbf{V}_2 = \mathbf{g} - (\mathbf{p} \cdot \mathbf{g})\mathbf{p}$ will be the projection of $\mathbf{g}$ onto the plane orthogonal to $\mathbf{p}$, as shown in figure 10.2.

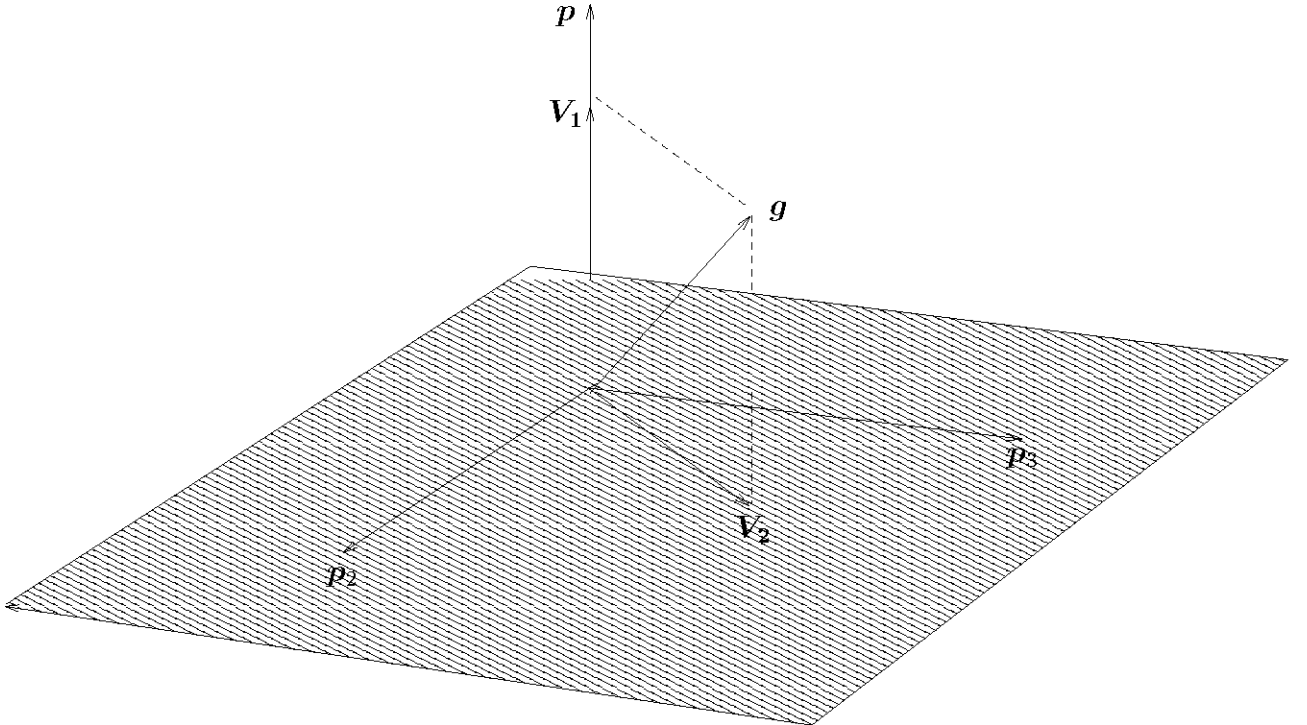


Figure 10.2: New coordinate system.

Let us consider a new orthonormal coordinate system with the first basis vector (first axis) equal to $\mathbf{p}$. In this new coordinate system vector $\mathbf{g}$ will have coordinates

$$\mathbf{g}' = (g'_1, \dots, g'_r) = V\mathbf{g}$$

obtained from $\mathbf{g}$ by orthogonal transformation

$$V = (\mathbf{p}, \mathbf{p}_2, \dots, \mathbf{p}_r)$$

that maps canonical basis into this new basis. But we proved in Lecure 4 that in that case $g'_1, \dots, g'_r$ will also be i.i.d. standard normal. From figure 10.2 it is obvious that vector $\mathbf{V}_2 = \mathbf{g} - (\mathbf{p} \cdot \mathbf{g})\mathbf{p}$ in the new coordinate system has coordinates

$$(0, g'_2, \dots, g'_r)^T$$

and, therefore,

$$|\mathbf{V}_2|^2 = |\mathbf{g} - (\mathbf{p} \cdot \mathbf{g})\mathbf{p}|^2 = (g'_2)^2 + \dots + (g'_r)^2.$$

But this last sum, by definition, has χ_{r-1}^2 distribution since $g'_2, \dots, g'_r$ are i.i.d. standard normal. This finishes the proof of Theorem. □

Chi-squared goodness-of-fit test for simple hypothesis.

Suppose that we observe an i.i.d. sample $X_1, \dots, X_n$ of random variables that take a finite number of values $B_1, \dots, B_r$ with unknown probabilities $p_1, \dots, p_r$. Consider hypotheses

$$\begin{aligned} H_0 : & \quad p_i = p_i^\circ \text{ for all } i = 1, \dots, r, \\ H_1 : & \quad \text{for some } i, p_i \neq p_i^\circ. \end{aligned}$$

If the null hypothesis H_0 is true then by Pearson's theorem

$$T = \sum_{i=1}^r \frac{(\nu_i - np_i^\circ)^2}{np_i^\circ} \rightarrow^d \chi_{r-1}^2$$

where $\nu_i = \#\{X_j : X_j = B_i\}$ are the observed counts in each category. On the other hand, if H_1 holds then for some index i , $p_i \neq p_i^\circ$ and the statistics T will behave differently. If p_i is the true probability $\mathbb{P}(X_1 = B_i)$ then by CLT

$$\frac{\nu_i - np_i}{\sqrt{np_i}} \rightarrow^d N(0, 1 - p_i).$$

If we rewrite

$$\frac{\nu_i - np_i^\circ}{\sqrt{np_i^\circ}} = \frac{\nu_i - np_i + n(p_i - p_i^\circ)}{\sqrt{np_i^\circ}} = \sqrt{\frac{p_i}{p_i^\circ}} \frac{\nu_i - np_i}{\sqrt{np_i}} + \sqrt{n} \frac{p_i - p_i^\circ}{\sqrt{p_i^\circ}}$$

then the first term converges to $N(0, (1 - p_i)p_i/p_i^\circ)$ and the second term diverges to plus or minus ∞ because $p_i \neq p_i^\circ$. Therefore,

$$\frac{(\nu_i - np_i^\circ)^2}{np_i^\circ} \rightarrow +\infty$$

which, obviously, implies that $T \rightarrow +\infty$. Therefore, as sample size n increases the distribution of T under null hypothesis H_0 will approach χ_{r-1}^2 -distribution and under alternative hypothesis H_1 it will shift to $+\infty$, as shown in figure 10.3.